

Overview and General Instructions for Use

The enclosed scripts compile and prepare two datasets of a visual working memory task for analysis.

*Scripts with the same numbering and title are redundant, meaning that a version is available in both R and Python. Python scripts written using Jupyter Notebooks and R scripts were written using R Studio.

Sources for the data

Dataset 1 contains two experiments from the following paper, referred to as Experiment 1 and Experiment 2:

Xu, Z., Adam, K. C. S., Fang, X., & Vogel, E. K. (2018). The reliability and stability of visual working memory capacity. *Behavior Research Methods*, 50(2), 576-588

The first experiment contains 135 participants who completed 540 trials in 9 blocks during a single session. The second experiment contains 79 participants who each completed 31 sessions (on days 1-30 and on day 60) of 120 trials each.

The data and corresponding documentation can be accessed here: <https://osf.io/g7txf/>

Dataset 2 contains one experiment and 3849 datasets of 120-130 trials in a single session from the following paper:

Balaban, H., Fukuda, K., & Luria, R. (2019). What can half a million change detection trials tell us about visual working memory? *Cognition*, 191, 103984

The data and corresponding documentation can be accessed here: <https://osf.io/mzs9e/>

Pipeline

Folder A_Preliminary

1. S0_Install_and_import_packages.R

Contains code to install and import all of the packages to be used in subsequent R scripts.

Folder B_Dataset_1

2. S1_Compile_Data_from_Experiment_1.m

This script extracts variables from each participant's MATLAB files within the shared Data_experiment_1/IndividualFiles/ and outputs these variables with one trial per row to a .csv file. The `cellflat.m` function must be in the main directory (**outside** of Data_experiment_1) prior to running the script. Additional stimulus properties are drawn from the MATLAB files that were not shared in previous compilations, including the item colors and locations in the entire memory array and at test for each trial.

3. S2_Check_expt_1_trial_datasets.R

This script compares overlapping columns from the csv file that was compiled by S1 with the compiled participant csv files that were shared by the authors in Data_experiment_1/CSV_Individuals/ to make sure the values match.

4. S3_Compile_Data_from_Experiment_2.m

This script extracts variables from each participant's MATLAB files within the shared Data_experiment_2/Matlab_individuals_E2/, and outputs these variables with one trial per row to a .csv file. The `cellflat.m` function must be in the main directory (**outside** of Data_experiment_2) prior to running the script. Additional stimulus properties are drawn from the MATLAB files that were not shared in previous compilations, including the item colors and locations in the entire memory array and at test for each trial.

5. S4_Check_expt_2_trial_datasets.R

This script compares overlapping columns from the csv file that was compiled by S3 with the compiled participant csv files that were shared by the authors in Data_experiment_2/CSV_individuals_E2/ to make sure the values match.

Folder C_Dataset_2

6. S5_Extract_and_combine_data.R/S5_Extract_and_combine_data.ipynb

This script compiles all datasets in subdirectories of the shared K folder into a single csv file. A `raw_trial_data.csv` is first output with the exact values from the raw text files, and then a final `Trial_Data.csv` is output that is formatted for the next script.

7. S6_Check_trial_datasets.R

This script checks that the csv file compiled by S5 in R and in Python are the same. To run this check, it is necessary to run the S5 Python script with the `R_File_List.txt` file created in the S5 R version so that the files will be compiled in the same order.

Folder D_Combined_Dataset

8. `S7_Combine_trial_datasets.R/S7_Combine_trial_datasets.ipynb`

This script does the following:

1. Dataset 1 is formatted for analysis and the following scripts are output:
 - a. `Dataset_expt1_formatted_data.csv`
 - b. `Dataset_expt2_formatted_data.csv`
2. Overlapping columns from these datasets are added to the dataset 2, and two combined datasets are created:
 - a. `Combined_trial_data_with_120_4or8_trials_per_individual_dataset.csv` (This scales the number of trials of set sizes 4 and 8 to be as close as possible to 120 for all datasets, with the first 160 trials per participant retained from dataset 1, experiment 1 and the first 168 trials per participant retained in dataset 1, experiment 2, plus all 120 trials per dataset in dataset 2).
 - b. `Combined_trial_data_with_all_trials_in_all_datasets.csv` (with all trials in both datasets included).

There are fewer columns in the combined files because the columns are dictated by what is present in dataset 2, which had fewer available variables compared to dataset 1.

Additional columns were added based on the stimulus properties described in the papers, including:

- `['study']` Which paper the data came from (1 = Xu et al., 2018, 2 = Balaban et al., 2019)
- `['experiment']` Experiment number within the paper
- `['session']` Which session, default is 1 where only one session was run
- `['distance_from_monitor']` in cm
- `['stim_size']` the size of a square in degrees of visual angle
- `['fixation_size']` the size of the fixation marker in degrees of visual angle
- `['stim_duration']` duration of the memory array presentation in ms
- `['intertrial_interval']` in ms
- `['retention_interval']` in ms
- `['break_duration']` duration of break in between blocks in ms